

RPX Benchmark: Comprehensive Metrics per Task

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Purpose: Exhaustive list of all standard evaluation metrics for each task in the RPX benchmark. Compute everything; select later for publication.

1. Monocular Depth Estimation

Currently implemented (`rpx_benchmark/metrics/depth.py`)

- AbsRel, RMSE, delta1, delta2, delta3

Full metric set to implement

Error metrics (lower is better)

Metric	Formula	Used by
AbsRel	$(1/N) \sum d_{\text{pred}_i} - d_i / d_i$	All papers
SqRel	$(1/N) \sum (d_{\text{pred}_i} - d_i)^2 / d_i$	Eigen et al. 2014, KITTI, DA-V2, Metric3D
RMSE	$\sqrt{(1/N) \sum (d_{\text{pred}_i} - d_i)^2}$	All papers
RMSElog	$\sqrt{(1/N) \sum (\log d_{\text{pred}_i} - \log d_i)^2}$	Eigen et al. 2014, NYU, DA-V2
SIlog (scale-invariant log)	$\sqrt{(1/N) \sum e_i^2 - (1/N^2 (\sum e_i)^2))} \times 100$, where $e_i = \log d_{\text{pred}_i} - \log d_i$	KITTI benchmark (primary ranking metric)
log10	$(1/N) \sum \log_{10} d_{\text{pred}_i} - \log_{10} d_i $	NYU v2, some indoor papers
iRMSE	RMSE computed on inverse depth (1/d)	KITTI depth prediction benchmark
iMAE	MAE computed on inverse depth (1/d)	KITTI depth prediction benchmark
MAE	$(1/N) \sum d_{\text{pred}_i} - d_i $	MDEC challenge, BenchDepth

Accuracy / threshold metrics (higher is better)

Metric	Formula	Used by
delta1 ($\delta < 1.25$)	% pixels where $\max(d_{\text{pred}}/d, d/d_{\text{pred}}) < 1.25$	All papers
delta2 ($\delta < 1.25^2$)	% pixels where $\max(d_{\text{pred}}/d, d/d_{\text{pred}}) < 1.5625$	All papers
delta3 ($\delta < 1.25^3$)	% pixels where $\max(d_{\text{pred}}/d, d/d_{\text{pred}}) < 1.953125$	All papers

Boundary / edge metrics (from MDEC and Depth Pro)

Metric	Formula / Description	Used by
F-score	Harmonic mean of accuracy and completeness at edge pixels	MDEC challenge (4th edition)
Accuracy (edges)	% predicted edges within threshold of GT edges	MDEC
Completeness (edges)	% GT edges that have a predicted edge nearby	MDEC

Metric	Formula / Description	Used by
ORD (ordinal error)	% pixel pairs where depth ordering is wrong	Some ordinal-focused papers

Alignment variants For relative-depth models (Marigold, Lotus, DA-V2 base), metrics must be computed after alignment:

Alignment	Description	When to use
None	Directly compare predicted vs GT (requires metric output)	UniDepth V2, Depth Pro, DA-V2 Metric, Metric3D v2, MoGe-2, MetricSolver, ZoeDepth
Median scaling	$\text{pred_aligned} = \text{pred} \times \text{median}(\text{GT}) / \text{median}(\text{pred})$	Legacy protocol
Least-squares (affine)	$\text{pred_aligned} = a \times \text{pred} + b$ (fit a,b by least squares)	MDEC 4th edition (new standard)
Least-squares in disparity	Same but fit on $1/d$	Some papers (Lotus, Marigold)

Per-region / stratified metrics

Stratification	Description
Near / Mid / Far	Split valid pixels by depth range (e.g., 01m, 13m, 3m+)
Object vs Background	Use segmentation masks to separate
Edge vs Interior	Canny/Sobel on GT depth to separate boundary pixels
Per-phase (RPX-specific)	Clutter / Interaction / Clean phase breakdown

Robotics-specific metrics

Metric	Description	Why it matters
Planarity error	RMSE of plane fit residuals on known-flat surfaces (table)	Grasp planning assumes flat support
Object-relative depth ordering	% object pairs with correct relative depth	Pick sequence planning
Depth discontinuity recall	% object boundaries where depth discontinuity is detected	Segmentation from depth

Reference for metric definitions: Eigen et al. “Depth Map Prediction from a Single Image using a Multi-Scale Deep Network”, NeurIPS 2014 (arXiv:1406.2283)

2. Instance Segmentation

Currently implemented (`rpx_benchmark/metrics/segmentation.py`)

- mIoU

Full metric set to implement

Metric	Formula / Description	Used by
mIoU	Mean IoU across GT classes	Standard
AP (Average Precision)	Area under PR curve at IoU=0.5	COCO
AP50	AP at IoU threshold 0.50	COCO, VOC
AP75	AP at IoU threshold 0.75	COCO
AP (COCO)	Mean AP at IoU thresholds [0.50:0.05:0.95]	COCO primary
AP(S) / AP(M) / AP(L)	AP for small / medium / large objects	COCO
PQ (Panoptic Quality)	Recognition quality x segmentation quality	Panoptic segmentation
SQ (Segmentation Quality)	Mean IoU of matched segments	Panoptic
RQ (Recognition Quality)	F1 of matched segments	Panoptic
Boundary IoU	IoU computed only at mask boundaries (within d pixels of edge)	Boundary-sensitive eval
Mask AP	AP using mask IoU rather than box IoU	COCO instance seg
F1 per class	Per-category F1 score	Balanced eval
FWIoU (Frequency Weighted IoU)	IoU weighted by class pixel frequency	Semantic seg
Pixel Accuracy	% correctly classified pixels	Semantic seg
Mean Accuracy	Mean per-class pixel accuracy	Semantic seg

RPX-specific stratification

- Per-phase (clutter / interaction / clean)
- Per-object-size (small < 32² px, medium < 96² px, large ≥ 96² px following COCO)
- Occluded vs non-occluded instances
- Known vs novel categories (for open-vocab setting)

3. Object Detection

Currently implemented (`rpx_benchmark/metrics/detection.py`)

- Precision, Recall, F1 at IoU=0.5

Full metric set to implement

Metric	Description	Used by
mAP	Mean AP across classes (IoU=0.5)	VOC
mAP@[.5:.95]	COCO-style mean AP at 10 IoU thresholds	COCO primary
AP50	AP at IoU=0.50	Standard
AP75	AP at IoU=0.75	COCO
AP(S) / AP(M) / AP(L)	AP by object size	COCO
AR_1 / AR_1_0 / AR_1_0_0	Average recall given 1 / 10 / 100 detections per image	COCO
Precision@IoU=0.5	Already implemented	
Recall@IoU=0.5	Already implemented	
F1@IoU=0.5	Already implemented	
mAP per category	Per-class AP breakdown	All
Localization error	Mean IoU of TP detections (how tight are the boxes)	Diagnosis

Metric	Description	Used by
Classification error rate	% TPs with wrong label	Diagnosis
Miss rate vs FPPI	At various false-positives-per-image rates	Pedestrian detection

4. Relative Camera Pose Estimation

Currently implemented (`rpx_benchmark/metrics/pose.py`)

- Rotation geodesic error (degrees), Translation L2 error (meters)

Full metric set to implement

Metric	Description	Used by
Rotation error (deg)	Geodesic: $\arccos((\text{tr}(\mathbf{R}_{\text{pred}}^T \mathbf{R}_{\text{gt}}) - 1) / 2)$	All pose papers
Translation error (m)	L2 norm of translation difference	All pose papers
Translation angular error (deg)	Angle between predicted and GT translation directions	MapFree, 7Scenes
AUC@5	Area under recall curve at 5 rotation threshold	FAR, RelPose
AUC@10	Area under recall curve at 10 rotation threshold	Standard
AUC@20	Area under recall curve at 20 rotation threshold	Standard
Median rotation error	Median across samples (robust to outliers)	Visual localization
Median translation error	Median across samples	Visual localization
Pose recall@(X, Y_m)	% pairs with $\text{rot} < X$ AND $\text{trans} < Y_m$	ScanNet, MapFree
Relative pose AUC	AUC of the cumulative recall vs threshold curve	8-point paper
Angular translation error	$\arccos(t_{\text{pred}} t_{\text{gt}} / (t_{\text{pred}} t_{\text{gt}}))$	When scale is ambiguous

Thresholded recall at standard cutoffs

- (5, 5cm), (5, 10cm), (10, 25cm), (15, 50cm) – indoor
- (5, 2m), (10, 5m) – outdoor

5. Keypoint Matching

Currently implemented (`rpx_benchmark/metrics/keypoints.py`)

- Keypoint accuracy (% within `px_threshold`), Mean match error

Full metric set to implement

Metric	Description	Used by
Match accuracy @3px	% correspondences within 3px of GT	ScanNet protocol
Match accuracy @5px	% within 5px	LoFTR, SuperGlue
Match accuracy @10px	% within 10px	Relaxed threshold
Mean matching error (px)	Already implemented	
Precision	$\text{TP} / (\text{TP} + \text{FP})$ for matches	SuperGlue
Recall	$\text{TP} / (\text{TP} + \text{FN})$ for matches	SuperGlue
Repeatability	% keypoints detected in both views	Feature detectors
Homography estimation AUC	AUC of corner reprojection error under estimated H	HPatches

Metric	Description	Used by
Pose estimation	From estimated E/F matrix	MegaDepth, ScanNet
AUC@5/10/20		
Number of inliers	After RANSAC	Practical measure
Inlier ratio	Inliers / total matches	Practical measure

6. Visual Grounding

Currently implemented (`rpx_benchmark/metrics/grounding.py`)

- Grounding IoU, Grounding accuracy@0.5

Full metric set to implement

Metric	Description	Used by
Acc@IoU=0.5	Already implemented	RefCOCO standard
Acc@IoU=0.25	Relaxed threshold	Some papers
Acc@IoU=0.75	Strict threshold	
Mean IoU	Average IoU of top prediction with GT	Already implemented
Top-5 accuracy	Correct box in top-5 predictions	Multi-box outputs
Pointing accuracy	Center of predicted box within GT box	Pointing game
GIOU (Generalized IoU)	IoU that accounts for non-overlapping boxes	Detection literature
Per-query-type accuracy	Breakdown by query complexity	Compositional
Unique vs Non-unique accuracy	How well it handles ambiguous references	RefCOCO+

7. Object Tracking

Currently implemented (`rpx_benchmark/metrics/tracking.py`)

- MOTA, IDF1, FP, FN, IDsw

Full metric set to implement

Metric	Description	Used by
MOTA	$1 - (FN + FP + IDsw) / GT$	MOT standard
MOTP	Mean IoU of matched detections	MOT standard
IDF1	ID F1-score (identity-aware)	MOT standard
HOTA	Higher Order Tracking Accuracy (geometric mean of detection + association)	MOT20/TAO
DetA	Detection accuracy component of HOTA	MOT20
AssA	Association accuracy component of HOTA	MOT20
MT (Mostly Tracked)	% GT tracks tracked for $\geq 80\%$ of lifetime	MOT
ML (Mostly Lost)	% GT tracks tracked for $\leq 20\%$ of lifetime	MOT
FP	Total false positives	Already output
FN	Total false negatives	Already output
IDsw	Total identity switches	Already output
Frag	Total fragmentations	MOT
Track mAP	AP treating tracking as detection + linking	TAO

8. Novel View Synthesis

Currently implemented (`rpx_benchmark/evaluators.py`)

- PSNR, SSIM (simplified global)

Full metric set to implement

Metric	Description	Used by
PSNR	Peak signal-to-noise ratio	All NVS papers
SSIM (structural, windowed)	Wang et al. 2004 – use sliding 11x11 window, NOT global	Standard; current impl is simplified
LPIPS	Learned perceptual similarity (AlexNet or VGG backbone)	All modern NVS papers
MS-SSIM	Multi-scale SSIM	Some papers
FID	Frchet Inception Distance (needs many samples)	Generative quality
DISTS	Deep image structure and texture similarity	Some papers
FLIP	HDR-VDP inspired perceptual metric	NVIDIA
AbsRel depth	If NVS also predicts depth, evaluate depth accuracy	NeRF-based methods
Masked PSNR/SSIM	Only on foreground (object) regions	Object-centric NVS

9. Sparse Depth

Currently implemented (`rpx_benchmark/metrics/sparse_depth.py`)

- Sparse AbsRel, Sparse RMSE

Full metric set to implement

Metric	Description	Used by
AbsRel (at sparse points)	Already implemented	
RMSE (at sparse points)	Already implemented	
MAE (at sparse points)	Mean absolute error at GT locations	Depth completion
iMAE	MAE on inverse depth	KITTI completion
iRMSE	RMSE on inverse depth	KITTI completion
delta1 at sparse points	Threshold accuracy at sparse locations	

10. Robot VQA (`rpx_benchmark/tasks/robot_vqa.py`)

Currently implemented

- MCQ accuracy, Per-task-type accuracy, Per-phase accuracy, Per-robot-operation accuracy

Full metric set to implement

Metric	Description	Used by
Overall accuracy	Already implemented	
Per task type (T1T6)	Already implemented	
Per phase	Already implemented	

Metric	Description	Used by
Per robot operation	Already implemented	
Confidence calibration	If model outputs logits: ECE (Expected Calibration Error)	VQA reliability
Abstention rate	% “cannot determine” answers	Safety-critical
STR (Scene Transition Robustness)	Accuracy change across phases	RPX-specific
Consistency	Same question across viewpoints -> same answer?	360 walkaround
Reference grounding accuracy	T1 specifically: separate TP/FP/TN/FN rates	Object registration
Spatial reasoning accuracy	T3 specifically: per-direction (L/R/above/below)	Spatial grounding

11. Cross-Task / Deployment Metrics

These apply to ALL tasks and are computed by the runner/profiler.

Three-Tier Resource Reporting (implemented in `rpx_benchmark.profiler`)

Tier 1 – Hardware-agnostic model properties (identical on any hardware):

Metric	Description	Status
Parameters (M)	Trainable parameter count	[YES] Implemented
FLOPs (G)	Forward-pass floating-point ops at batch=1, FP32	[YES] Implemented
MACs (G)	Multiply-accumulate ops (= FLOPs / 2)	[YES] Implemented
Memory Traffic (GB)	Estimated DRAM read+write bytes	[YES] Implemented
Arithmetic Intensity (FLOP/Byte)	FLOPs / Traffic – determines compute- vs memory-bound	[YES] Implemented

Tier 2 – Roofline latency bounds (hardware-parametric, anyone can recompute):

Metric	Description	Status
Compute-bound time (ms)	FLOPs / GPU peak FLOP/s	[YES] Implemented
Memory-bound time (ms)	Traffic / GPU peak bandwidth	[YES] Implemented
Roofline latency (ms)	max(compute, memory) – theoretical lower bound	[YES] Implemented
Bottleneck	“compute” or “memory”	[YES] Implemented

Published for three reference GPUs: A100-80GB, RTX 4090, Jetson Orin 64GB. Custom GPUs supported via GPUSpec.

Tier 3 – Measured (hardware-specific, with SystemCard):

Metric	Description	Status
Latency (p50, p95, p99, mean)	Per-sample inference time	[YES] Implemented

Metric	Description	Status
Peak memory (CPU, CUDA, MPS)	Max memory usage	[YES] Implemented
SystemCard	GPU name, VRAM, CPU, RAM, CUDA version, PyTorch version, OS, precision	[YES] Implemented
Throughput (FPS)	Frames per second at batch=1	Derivable from latency
Batch throughput	FPS at batch=4, 8, 16	Add to profiler

Composite + Deployment Metrics

Metric	Description	Status
Deployment Readiness Score (DRS)	Platform-independent: TP x R x E. TP = task accuracy [0,1]; R = 1- STR [0,1]; E = 1/(1 + FLOPs/F_median) [0,1]. Multiplicative – zero in any component kills the score. Median-anchored efficiency means no arbitrary budgets.	[YES] Implemented
Embodied Readiness Score (ERS)	Legacy composite of accuracy + robustness + latency + memory + compute (hardware-dependent)	[YES] Implemented (superseded by DRS for paper)
Temporal Stability	Consistency across consecutive frames	[YES] Implemented (depth, seg)
Geometric Coherence (SGC)	Depth <-> segmentation consistency	[YES] Implemented (seg)
TensorRT/ONNX compatibility	Binary: can the model be exported?	Manual flag
Warm-up frames	How many frames until stable predictions	Temporal models

Implementation Priority

Phase 1 (immediate – needed for depth benchmark): Extend `depth.py` with SqRel, RMSElog, SIlog, log10, MAE, iRMSE, iMAE, alignment variants, and per-region stratification.

Phase 2 (before submission): Full COCO-style detection/segmentation AP, windowed SSIM, LPIPS for NVS, HOTA for tracking, full pose recall tables.

Phase 3 (nice-to-have): Boundary metrics (F-score, Boundary IoU), planarity error, FID, calibration metrics.

References

1. Eigen et al. “Depth Map Prediction from a Single Image using a Multi-Scale Deep Network.” NeurIPS 2014. arXiv:1406.2283
2. KITTI Depth Prediction Benchmark: https://cvlibs.net/datasets/kitti/eval_depth.php
3. “How Should One Evaluate Monocular Depth Estimation?” arXiv:2510.19814
4. 4th MDEC Challenge. arXiv:2504.17787
5. BenchDepth. arXiv:2507.15321
6. COCO Evaluation Metrics: <https://cocodataset.org/#detection-eval>
7. MOT Metrics: <https://motchallenge.net/>
8. HOTA: Luiten et al. “HOTA: A Higher Order Metric for Evaluating Multi-Object Tracking.” IJCV 2021.